PROJECT DEVELOPMENT PHASE

DATE	02 March 2026
TEAM ID	LTVIP2026TMIDS28461
PROJECT TITLE	Visualization Tool for Electric Vehicle Charge and Range Analysis
MAXIMUM MARKS	15 Marks

1. Data Connectivity

The data required for this project is collected from publicly available sources and stored in CSV file format. The datasets include electric vehicle specifications and charging station details. These CSV files are directly connected to Tableau for visualization and analysis.

Links of Data Sets:

https://drive.google.com/drive/folders/1Rkzdks6Us1Uq2SRB4nxMAb83jN5bpHll

2. Data Preparation

Data preparation is an important step to ensure accuracy and consistency. The following preprocessing steps were performed before visualization:

- Removed missing and null values
- Removed duplicate records
- Renamed column headers for better understanding
- Converted units such as range(km)and battery capacity(kWh)into standard formats
- Filtered unnecessary attributes
- Checked data consistency and corrected errors

After pre-processing, the cleaned data was used for creating dashboards and stories.

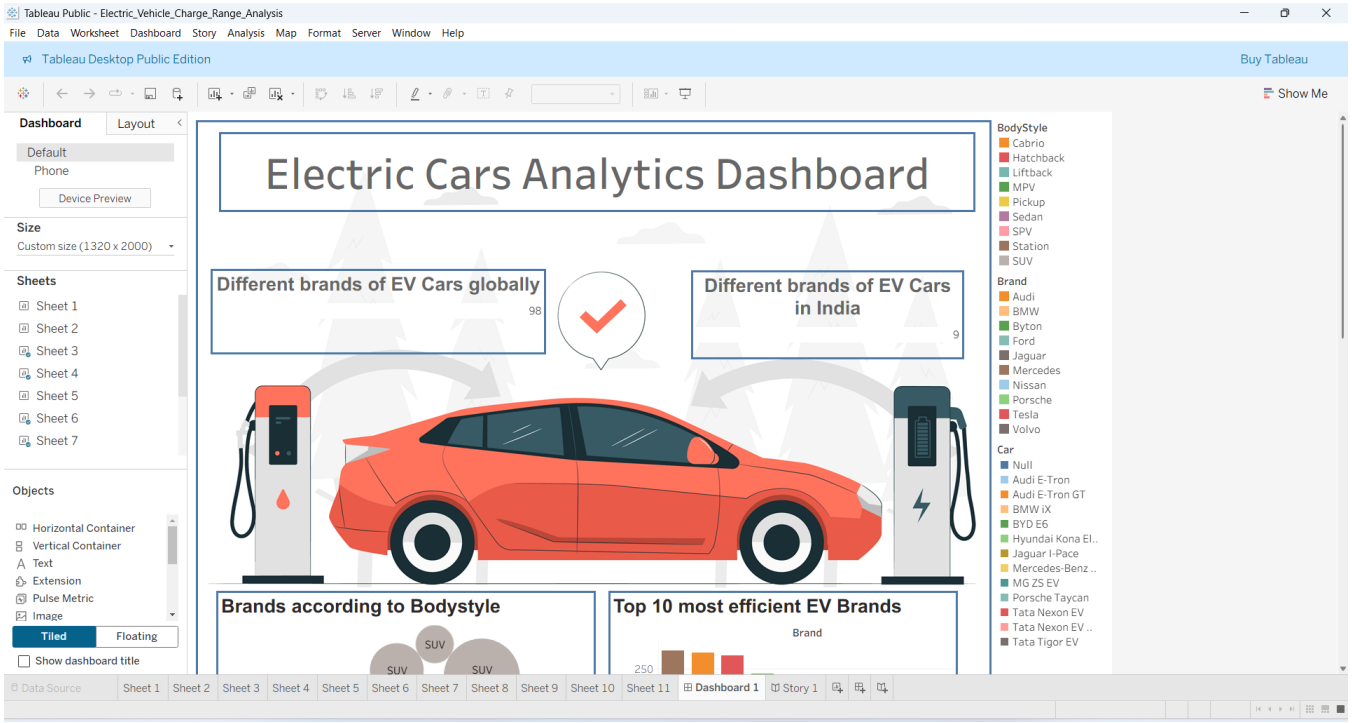
3. Dashboard

A Tableau dashboard was created by combining multiple visualizations to provide a clear understanding of electric vehicle charge and range analysis. The dashboard includes:

- Comparison of EV range across different models
- Analysis of battery capacity vs charging time
- Distribution of charging stations using map visualization
- Price vs range comparison

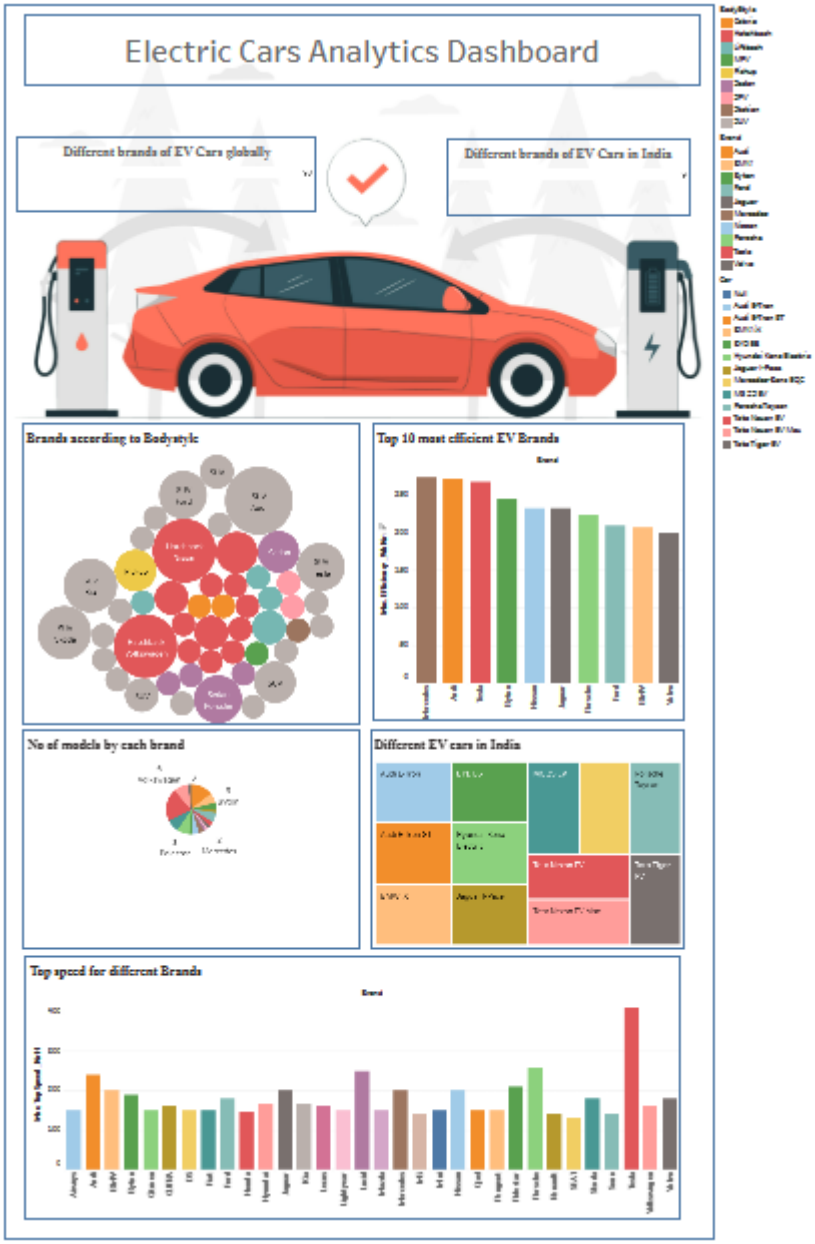
The dashboard is interactive and allows users to filter data based on EV model, brand, and range.

Screenshots :-



Dashboard in Tableau Public Link :-

https://public.tableau.com/views/Electric_Vehicle_Charge_Range_Analysis/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link



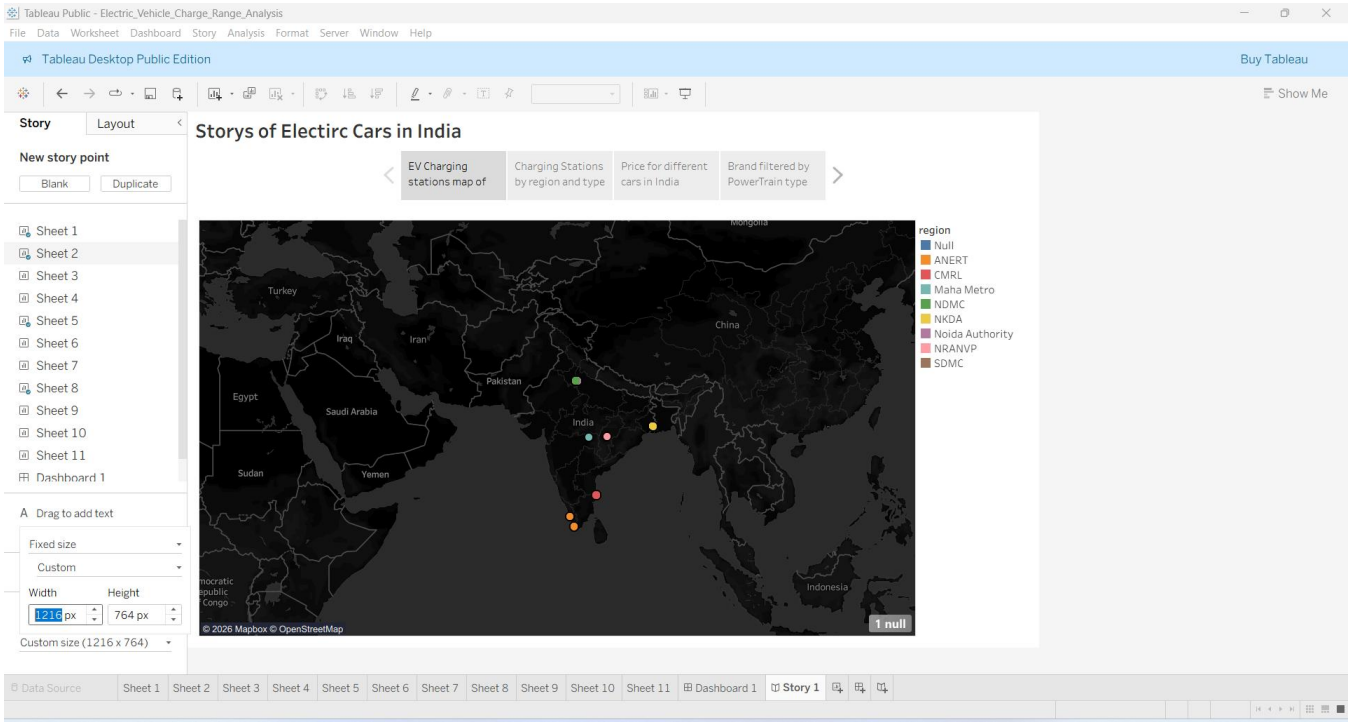
4. Story

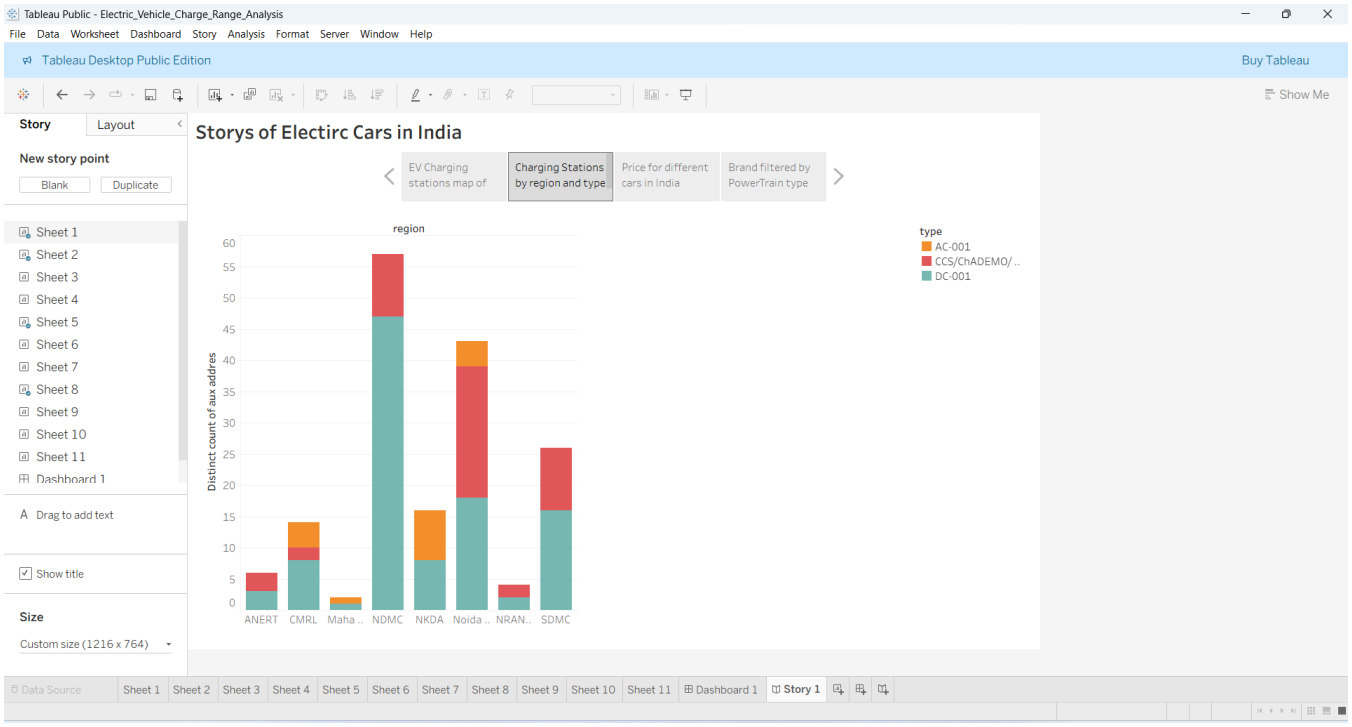
A Tableau story was created to present insights in a step-by-step manner. The story consists of multiple story points that guide the user through the analysis:

1. Overview of electric vehicle range
2. Comparison of charging time and battery capacity
3. Availability of charging stations
4. Key insights and observations

The story helps users easily understand trends and make in formed decisions.

Screenshots :-





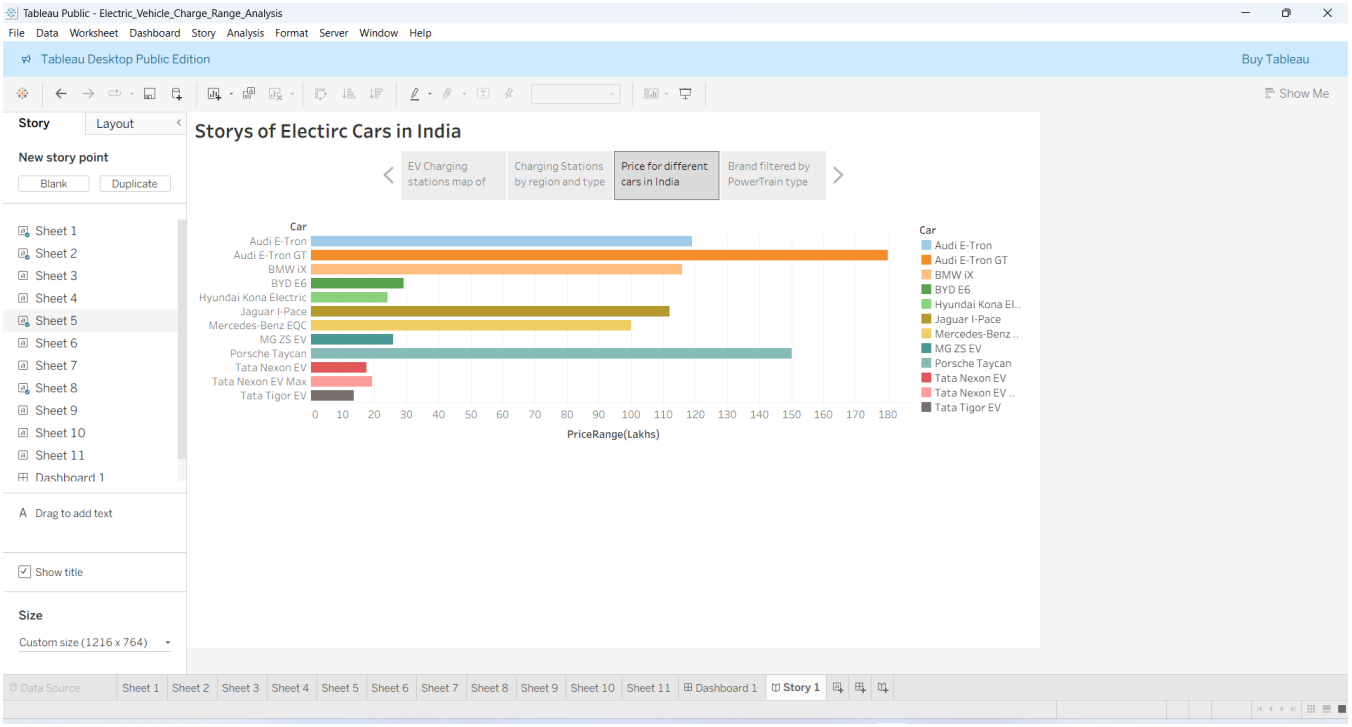
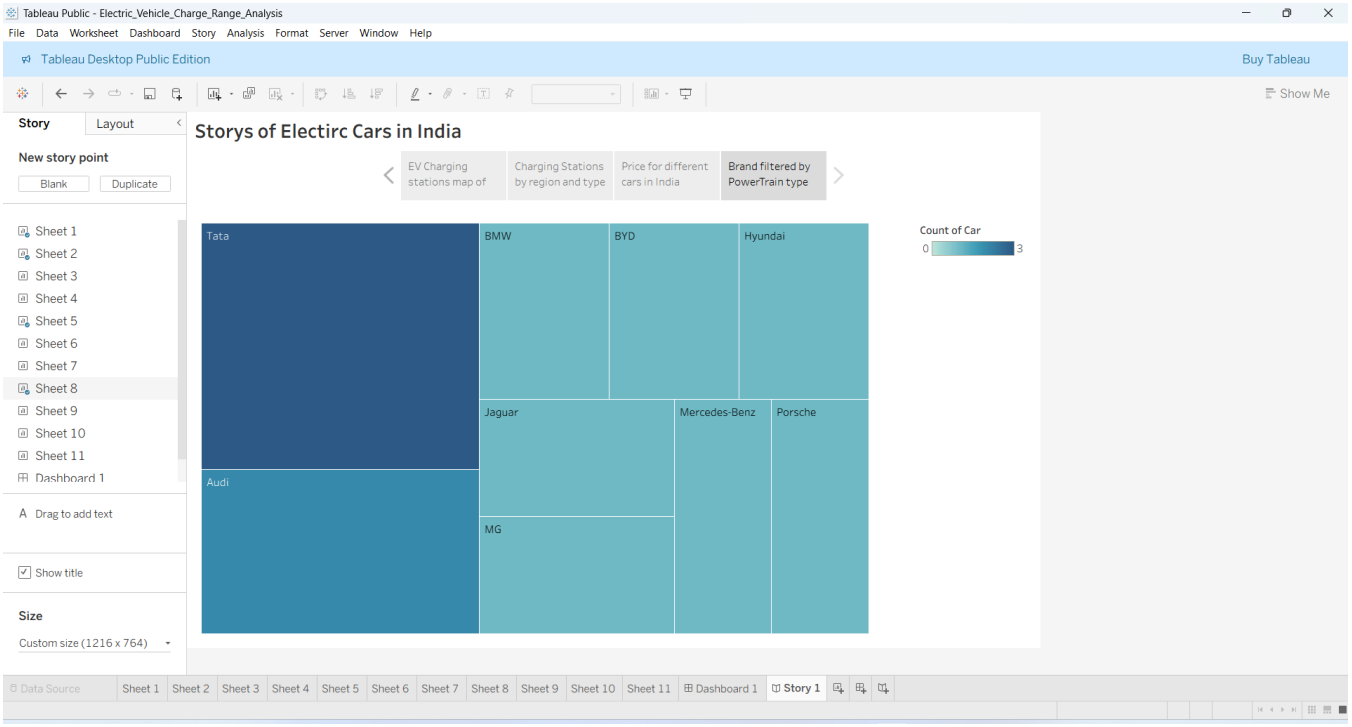


Tableau Public - Electric_Vehicle_Charge_Range_Analysis
File Data Worksheet Dashboard Story Analysis Format Server Window Help
Tableau Desktop Public Edition
Buy Tableau
Show Me
Story
Layout
New story point
Blank
Duplicate
Sheet 1
Sheet 2
Sheet 3
Sheet 4
Sheet 5
Sheet 6
Sheet 7
Sheet 8
Sheet 9
Sheet 10
Sheet 11
Dashboard 1
Drag to add text
Show title
Size
Custom size (1216 x 764)
Storys of Electirc Cars in India
EV Charging stations map of
Charging Stations by region and type
Price for different cars in India
Brand filtered by PowerTrain type
Car
Audi E-Tron
Audi E-Tron GT
BMW iX
BYD E6
Hyundai Kona Electric
Jaguar I-Pace
Mercedes-Benz EQC
MG ZS EV
Porsche Taycan
Tata Nexon EV
Tata Nexon EV Max
Tata Tigor EV
0 10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180
PriceRange(Lakhs)
Hyundai Kona El..
Mercedes-Benz ..
Tata Nexon EV ..
Data Source
Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Sheet 9 Sheet 10 Sheet 11 Dashboard 1 Story 1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MG
Count of Car
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3
Data Source
Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8 Sheet 9 Sheet 10 Sheet 11 Dashboard 1 Story 1

Story board in Tableau Public :-

https://public.tableau.com/views/Electric_Vehicle_Charge_Range_Analysis/Story1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

5. Creativity(Font and Style)

The project document is designed using a clean and professional layout. Consistent fonts, proper headings, and structured sections are used throughout the document. Visual elements such as charts and dashboards are clearly labeled and aligned. The overall design improves readability and makes the project easy to understand.

- Applied clear headings and sub headings to separate sections
- Used bullet points instead of long paragraphs to improve clarity
- Maintained proper spacing and alignment across all pages
- Used meaningful titles and captions for dashboards and stories
- Applied color the in dashboard story visually differentiate data categories